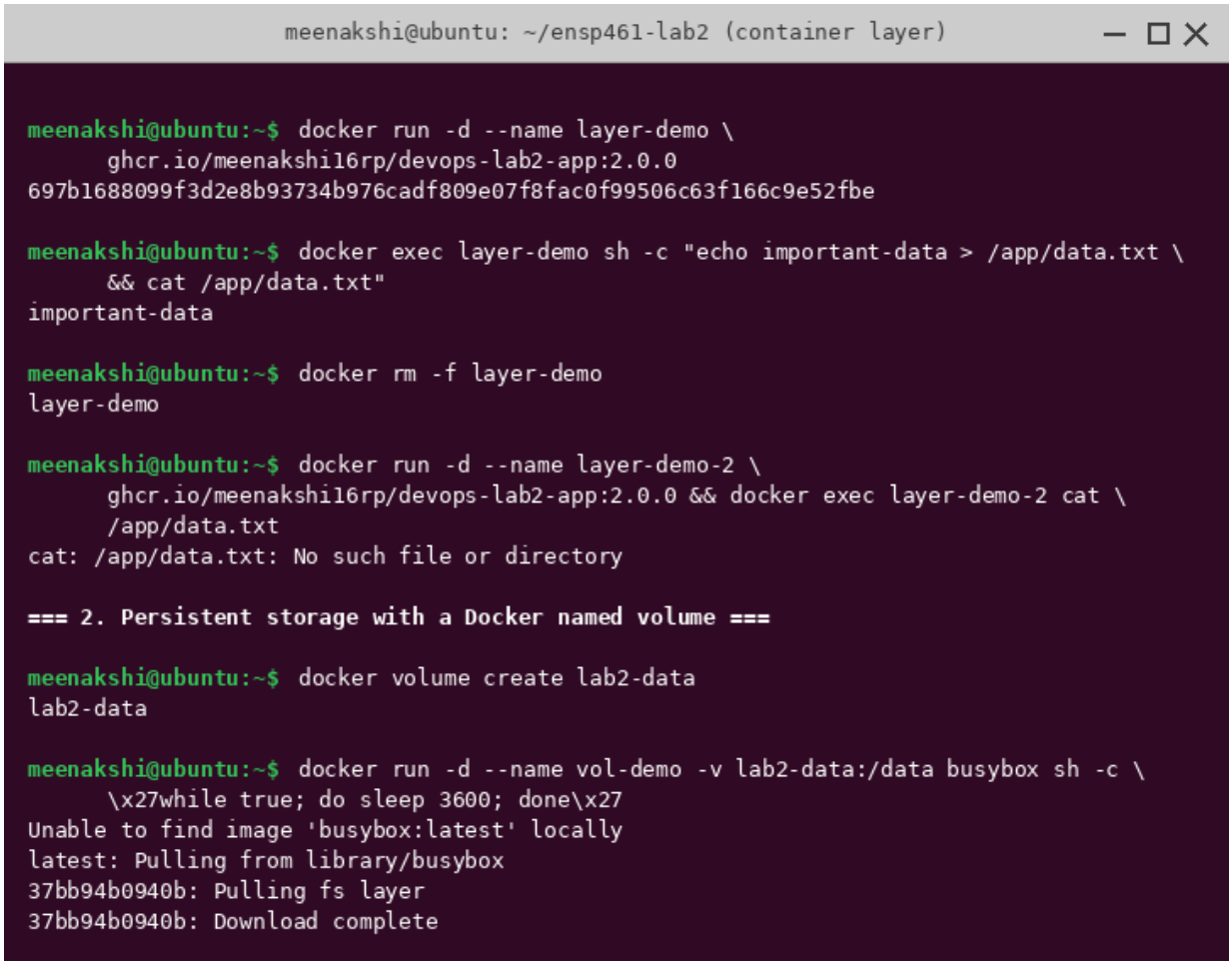
Question 3: Demonstrate loss of container-layer data after container deletion, and permanent storage using a Docker named volume and a bind mount.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. Container-layer data is lost when the container is deleted



2. Permanent storage with a Docker named volume

```
meenakshi@ubuntu: ~/ensp461-lab2 (volume + bind mount)

37bb94b0940b: Pull complete
Digest: sha256:a46257e06c82ad85242d0d38077d0a52d9896389ca60a1a59d3463db586e5fa4
Status: Downloaded newer image for busybox:latest
d5b57a9d3968ee0aa5c427002a31286da99f33bc3222ed60aee1a4c03a68c244

meenakshi@ubuntu:~$ docker exec vol-demo sh -c "echo persistent-data > /data/note.txt \
      && cat /data/note.txt"
persistent-data

meenakshi@ubuntu:~$ docker rm -f vol-demo
vol-demo

meenakshi@ubuntu:~$ docker run --rm -v lab2-data:/data busybox cat /data/note.txt
persistent-data
```

volume details

```
meenakshi@ubuntu:~$ docker volume create lab2-data
lab2-data
meenakshi@ubuntu:~$ docker volume inspect lab2-data --format {{.Mountpoint}}
/var/lib/docker/volumes/lab2-data/_data
```

3. Permanent storage with a bind mount

```
meenakshi@ubuntu: ~/ensp461-lab2 (bind mount)

=== 3. Permanent storage with a bind mount ===

meenakshi@ubuntu:~$ mkdir -p ~/bindmount-demo && echo host-file-content > \
      ~/bindmount-demo/host.txt

meenakshi@ubuntu:~$ docker run --rm -v /home/meenakshi/bindmount-demo:/hostdata busybox \
      cat /hostdata/host.txt
host-file-content

meenakshi@ubuntu:~$ docker run --rm -v /home/meenakshi/bindmount-demo:/hostdata busybox \
      sh -c "echo written-by-container > /hostdata/from-container.txt"

meenakshi@ubuntu:~$ cat ~/bindmount-demo/from-container.txt   # written by the \
      container, read on the host
written-by-container
```

4. Observations

- Files written to the container's writable layer disappear with the container (/app/data.txt gone in a fresh container) - container layers are ephemeral.
- The named volume survived container deletion: a brand-new container mounting lab2-data read the same /data/note.txt.
- The bind mount maps a host directory directly: the host file was readable inside the container and a file written by the container appeared immediately on the host.
- Rule of thumb: named volumes for container-managed data, bind mounts for host development and configuration.